AI IN HEALTHCARE · HIGH-RISK PROJECT PROPOSAL

VLM-Powered Concept Bottleneck Models for Dermatology,

with a Fitzpatrick Skin-Tone Fairness Audit

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1. Motivation

Skin-cancer classifiers are trained predominantly on light-skinned cohorts, and they under-perform on darker skin. The disparity is documented (Daneshjou et al., 2022) showed a 10–15-point AUROC drop on Fitzpatrick V–VI imagery in DDI but existing audits stop at the diagnosis-level number. They do not tell a clinician or model developer which clinical feature is being mis-detected, only that the overall score is lower. Post-hoc saliency tools such as Grad-CAM surface where the model looked, not what it understood. This limits both trust and remediation which is a fairness gap becomes a dataset-collection moonshot rather than a targeted feature-engineering task.

Concept Bottleneck Models (Koh et al., 2020) route every prediction through an intermediate layer of human-interpretable concepts, and they can be audited and intervened upon directly. Recent work has begun to apply vision-language models (VLMs) to populate the concept layer without per-image concept labels (Patrício et al., 2024). What is still missing is the combination of

(i) a VLM-powered CBM in dermatology with

(ii) a Fitzpatrick-stratified audit of both the concept layer and the diagnosis layer,

(iii) the quantitative probes such as concept-intervention rescue and faithfulness

That is the gap this project targets.

2. Research questions

a. Can an interpretable concept-driven head match a deep CNN baseline on dermoscopic melanoma classification?

b. Do concept-prediction errors concentrate in specific Fitzpatrick skin-tone buckets, and how does the per-bucket diagnosis gap compare against opaque baselines?

c. Do concept interventions rescue misdiagnoses, and which concepts carry the largest corrective signal?

d. Are CBM explanations quantitatively more faithful than Grad-CAM on the same images?

3. Proposed approach

The pipeline has five stages plus one bonus faithfulness stage.

a. Stage 1 uses a frozen BiomedCLIP (Zhang et al., 2023) to score nine dermoscopic concepts drawn from the ABCDE rule and the 7-point checklist which is the pigment network, dots/globules, blue-white veil, streaks, regression, vessels, asymmetry, border, colour variation via softmax over positive/negative prompt pairs.

b. Stage 2 trains a sparse L1-regularised logistic-regression head on the nine concept probabilities (5-fold stratified CV on Derm7pt), with an MLP head as a non-linear reference.

c. Stage 3 fine-tunes EfficientNet-B0 and ResNet-50 on HAM10000 as opaque baselines equipped with Grad-CAM.

d. Stage 4 is the audit with metrics like per-concept AUROC, per-bucket AUROC / sensitivity / specificity / ECE, and equalized-odds and demographic-parity gaps are calculated with bootstrap 95 % confidence intervals.

e. Stage 5 simulates a concept-intervention oracle that replaces one VLM concept with its ground-truth value and re-runs the head, reporting rescue rates per concept.

f. Stage 6 applies Petsiuk et al.'s (2018) RISE deletion/insertion metric to CBM concepts and to Grad-CAM pixels, side-by-side.

4. Datasets

Four cohorts participate, each with a distinct role.

Dataset	Size	Role in the audit	Skin-tone coverage
HAM10000	10,015 images	Training data for EfficientNet-B0 and ResNet-50 baselines	Unlabelled and assumed light-skin skew
Derm7pt	1,011 (294 pos)	CBM cross-validation, per-concept AUROC	No Fitzpatrick labels
DDI	656 (171 pos)	Primary fairness audit using expert Fitzpatrick labels	Light I–II · Med III–IV · Dark V–VI
Fitzpatrick17k	~16,577 CSV	Pooled per-concept × per-bucket heatmap	Full 6-bucket coverage

5. Deliverables & expected outcomes

The project yields the following technical artefacts:

a. Per-bucket fairness tables with bootstrap 95 % Cis: Per-bucket AUROC, sensitivity, specificity and ECE on DDI, with 1000-resample percentile CIs, comparing CBM-LR against EfficientNet-B0 and ResNet-50.

b. Concept-count ablation: CBM-LR retrained at $k \in \{3, 5, 7, 9\}$ to quantify how few concepts are actually needed at the head.

c. **Concept-intervention rescue analysis:** Per-concept rescue rate and greedy top-k correction rate, localizing the diagnostic error budget to specific clinical features.

d. **Deletion/insertion faithfulness comparison:** Side-by-side AUCs for CBM explanations against Grad-CAM, testing whether the concept layer carries decision signal that saliency maps cannot.

The expected qualitative finding is that a CBM head matches the opaque baseline on overall AUROC while exposing an addressable, concept-level failure mode on darker skin. Even if the concept-layer gap is comparable to the diagnosis-layer gap in magnitude, localizing it to a small number of concepts (our running hypothesis is blue-white veil and vessels) turns remediation into a targeted concept-retraining task rather than a dataset-collection moonshot.

6. Scope, risks, and why this counts as high-risk

The project is high-risk along three axes:

a. First, zero-shot VLM concept prediction is noisy as BiomedCLIP may simply fail to pick up several concepts, in which case the CBM head has nothing to learn from.

b. Second, DDI's dark-skin subset is small ($n \approx 100$), which means the key fairness claim rests on a cohort that naturally produces wide bootstrap intervals; a negative or inconclusive result is a plausible outcome.

c. Third, Fitzpatrick17k images require a separate web scrape. The pipeline is designed to auto-skip bundles whose concept-score range collapses to noise, so the system degrades gracefully to DDI-only but this does constrain the pooled audit we would ideally run.